

**Praktikum  
STUDI KASUS 2**

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## 1. Kode:

Class Pembeli:

```
package JobsheetCM2Edit.Model;
```

```
public class Pembeli10 {
    public String namaPembeli;
    public String noHp;
    public int noAntri;
    public boolean isPrio;

    public Pembeli10(int noAntri, String nama, String no, boolean isPrio) {
        namaPembeli = nama;
        noHp = no;
        this.noAntri = noAntri;
        this.isPrio = isPrio;
    }

    public void tampilPembeli() {
        System.out.printf("%-15d %-10s %-12s\n", noAntri, namaPembeli, noHp);
    }
}
```

Class Pesanan:

```
package JobsheetCM2Edit.Model;
```

```
public class Pesanan10 {
    public int kodePesanan;
    public String namaPesanan;
    public int harga;

    public Pesanan10(int kode, String namaPesanan, int harga) {
        this.kodePesanan = kode;
        this.namaPesanan = namaPesanan;
        this.harga = harga;
    }
}
```

Class Node10:

```
package JobsheetCM2Edit.Model;
```

```
public class Node10 {
    public Pembeli10 dataPembeli;
    public Pesanan10 dataPesanan;
    public Node10 prev;
```

```

    public Node10 next;

    public Node10() {}

    public Node10(Pembeli10 dataPembeli) {
        this.dataPembeli = dataPembeli;
        this.prev = null;
        this.next = null;
    }

    public Node10(Pesanan10 dataPesanan) {
        this.dataPesanan = dataPesanan;
    }
}

```

Class AntrianPembeli:

```

package JobsheetCM2Edit.Model;

public class AntrianPembeli10 {
    public Node10 head;
    public Node10 tail;

    public AntrianPembeli10() {
        head = null;
        tail = null;
    }

    public boolean isEmpty() {
        return head == null;
    }

    public void addLast(Pembeli10 data, boolean isPrio) {
        Node10 newNode = new Node10(data);
        if (isEmpty()) {
            head = tail = newNode;
        } else {
            if (isPrio) {
                if (head.next != null && head.next.next != null) {
                    newNode.next = head.next.next;
                    if (head.next.next != null) {
                        head.next.next.prev = newNode;
                    }
                    newNode.prev = head.next;
                    head.next.next = newNode;
                    newNode.dataPembeli.noAntri = 3;
                }
            }
        }
    }
}

```

```

    } else {
        newNode.next = head.next;
        newNode.prev = head;
        if (head.next != null) {
            head.next.prev = newNode;
        }
        head.next = newNode;
        newNode.dataPembeli.noAntri = 2;
    }

    Node10 current = newNode.next;
    while (current != null) {
        current.dataPembeli.noAntri += 1;
        current = current.next;
    }
} else {
    tail.next = newNode;
    newNode.prev = tail;
    tail = newNode;
}
}
}

public void print() {
    if (isEmpty()) {
        System.out.println("Antrian masih kosong\n");
        return;
    }

    Node10 current = head;

    while (current != null) {
        Pembeli10 currentPembeli = current.dataPembeli;
        System.out.printf("%-15d %-10s %-12s\n", currentPembeli.noAntri,
currentPembeli.namaPembeli,
        currentPembeli.noHp);
        current = current.next;
    }
}

public void removeFirst() {
    if (isEmpty()) {
        System.out.println("Antrian masih kosong.\n");
        return;
    }
}

```

```

        if (head == tail) {
            head = tail = null;
            System.out.println("Data Antrian Pembeli berhasil dihapus.");
        } else {
            head = head.next;
            head.prev = null;
            System.out.println("Data Antrian Pembeli berhasil dihapus.");
        }
    }
}

```

Class AntrianPesanan:

```

package JobsheetCM2Edit.Model;

public class AntrianPesanan10 {
    public Node10 head;
    public Node10 tail;

    public AntrianPesanan10() {
        head = null;
        tail = null;
    }

    public boolean isEmpty() {
        return head == null;
    }

    public void addLast(Pesanan10 data) {
        Node10 newNode = new Node10(data);
        if (isEmpty()) {
            head = tail = newNode;
        } else {
            tail.next = newNode;
            newNode.prev = tail;
            tail = newNode;
        }
    }

    public void print() {
        if (isEmpty()) {
            System.out.println( "Antrian masih kosong\n");
            return;
        }

        Node10 current = head;
    }
}

```

```

        sortByNamaPesanan();

        while (current != null) {
            Pesanan10 currentPesanan = current.dataPesanan;
            System.out.printf("%-15d %-15s %-10d\n", currentPesanan.kodePesanan,
currentPesanan.namaPesanan, currentPesanan.harga);
            current = current.next;
        }
    }

    public void removeFirst() {
        if (isEmpty()) {
            System.out.println("Antrian masih kosong.\n");
            return;
        }

        if (head == tail) {
            head = tail = null;
            System.out.println("Data Antrian berhasil dihapus.\n");
        } else {
            head = head.next;
            head.prev = null;
            System.out.println("Data Antrian berhasil dihapus.\n");
        }
    }

    public void sortByNamaPesanan() {
        Node10 current = head;
        Node10 currNext = null;
        Pesanan10 temp;

        while (current != null) {
            currNext = current.next;

            while (currNext != null) {
                int resultCompare =
current.dataPesanan.namaPesanan.compareTo(currNext.dataPesanan.namaPesanan);

                if (resultCompare > 0) {
                    temp = current.dataPesanan;
                    current.dataPesanan = currNext.dataPesanan;
                    currNext.dataPesanan = temp;
                }

                currNext = currNext.next;
            }
        }
    }

```

```

    }
    current = current.next;
  }
}
}

```

Class View:

```

package JobsheetCM2Edit.View;

import java.util.Scanner;

import JobsheetCM2Edit.Model.*;

public class View10 {
    public int cetakMenu(Scanner sc, int pil) {
        System.out.println("=====");
        System.out.println("SISTEM ANTRIAN ROYAL DELISH");
        System.out.println("=====");
        System.out.println("1. Tambah Antrian");
        System.out.println("2. Cetak Antrian");
        System.out.println("3. Hapus Antrian dan Pesan");
        System.out.println("4. Laporan Pesanan");
        System.out.println("0. Keluar");
        System.out.print("Pilih menu: ");
        pil = sc.nextInt();
        sc.nextLine();
        return pil;
    }

    int noAntri = 1;
    public void tambahAntriPembeli(Scanner sc, AntrianPembeli10 antrian) {
        System.out.print("Nama Pembeli : ");
        String nmPembeli = sc.nextLine();
        System.out.print("No HP : ");
        String no = sc.nextLine();

        // Antrian prioritas
        System.out.print("Apakah antrian prioritas? (y/n): ");
        String prioritas = sc.nextLine();

        Pembeli10 pembeli;
        if (prioritas.equalsIgnoreCase("y")) {
            pembeli = new Pembeli10(noAntri, nmPembeli, no, true);
            antrian.addLast(pembeli, true);

```

```

    } else {
        pembeli = new Pembeli10(noAntri, nmPembeli, no, false);
        antrian.addLast(pembeli, false);
    }

    System.out.println("Antrian berhasil ditambahkan dengan nomor: " +
pembeli.noAntri + "\n");
    noAntri++;
}

public void tambahAntriPesanan(Scanner sc, AntrianPesanan10 antrian) {
    System.out.println("=== TAMBAH PESANAN ===");
    System.out.print("Kode Pesanan : ");
    int kd = sc.nextInt();
    sc.nextLine();
    System.out.print("Nama Pesanan : ");
    String nm = sc.nextLine();
    System.out.print("Harga      : ");
    int harga = sc.nextInt();
    sc.nextLine();
    Pesanan10 pesanan = new Pesanan10(kd, nm, harga);
    antrian.addLast(pesanan);
}

public void cetakAntrian(AntrianPembeli10 antrian) {
    System.out.println("=====");
    System.out.println("Daftar Antrian Pembeli");
    System.out.println("=====");

    if (!antrian.isEmpty()) {
        System.out.printf("%-15s %-10s %-12s\n", "No Antrian", "Nama", "No HP");
        antrian.print();
        System.out.println();
    } else {
        System.out.println("Antrian masih kosong.\n");
    }
}

public void hapusNPesan(Scanner sc, AntrianPembeli10 antrianPembeli,
AntrianPesanan10 antrianPesanan) {
    if (antrianPembeli.isEmpty()) {
        System.out.println("Antrian masih kosong.\n");
    }

    String nmPembeli = antrianPembeli.head.dataPembeli.namaPembeli;

```



```

        antrianPembeli.removeFirst();

        tambahAntriPesanan(sc, antrianPesanan);
        String nmPesanan = antrianPesanan.tail.dataPesanan.namaPesanan;
        System.out.println(nmPembeli + " telah memesan " + nmPesanan + ".\n");
    }

    public void laporanPesanan(AntrianPesanan10 antrian) {
        System.out.println("=====");
        System.out.println("LAPORAN PESANAN (URUT NAMA PESANAN)");
        System.out.println("=====");

        if (!antrian.isEmpty() && antrian.head.dataPesanan != null) {
            System.out.printf("%-15s %-15s %-10s\n", "Kode Pesanan", "Nama Pesanan",
"Harga");
            antrian.print();
            System.out.println();
        } else {
            System.out.println("Laporan masih kosong.\n");
        }
    }
}

```

Class SistemAntrianKasir:  
package JobsheetCM2Edit;

```

import java.util.Scanner;
import JobsheetCM2Edit.View.*;
import JobsheetCM2Edit.Model.*;

```

```

public class SistemAntrianKasir10 {

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        AntrianPembeli10 antrianBeli= new AntrianPembeli10();
        AntrianPesanan10 antrianPesan = new AntrianPesanan10();
        View10 view = new View10();

        int pil = 0;
        do {
            pil = view.cetakMenu(sc, pil);

            switch (pil) {

```

```
        case 1:
            view.tambahAntriPembeli(sc, antrianBeli);
            break;

        case 2:
            view.cetakAntrian(antrianBeli);
            break;

        case 3:
            view.hapusNPesan(sc, antrianBeli, antrianPesan);
            break;

        case 4:
            view.laporanPesanan(antrianPesan);
            break;

        case 0:
            System.out.println("Keluar.");
            break;

        default:
            System.out.println("Pilihan tidak valid.\n");
            break;
    }
} while (pil != 0);

sc.close();
}
```

## 2. Output:

```
=====
SISTEM ANTRIAN ROYAL DELISH
=====
1. Tambah Antrian
2. Cetak Antrian
3. Hapus Antrian dan Pesan
4. Laporan Pesanan
0. Keluar
Pilih menu: 1
Nama Pembeli : t
No HP : 0825
Apakah antrian prioritas? (y/n): n
Antrian berhasil ditambahkan dengan nomor: 5

=====
SISTEM ANTRIAN ROYAL DELISH
=====
1. Tambah Antrian
2. Cetak Antrian
3. Hapus Antrian dan Pesan
4. Laporan Pesanan
0. Keluar
Pilih menu: 2

=====
Daftar Antrian Pembeli
=====
No Antrian Nama No HP
1 q 0821
2 w 0822
3 e 0823
4 r 0824
5 t 0825
```

```
=====
SISTEM ANTRIAN ROYAL DELISH
=====
1. Tambah Antrian
2. Cetak Antrian
3. Hapus Antrian dan Pesan
4. Laporan Pesanan
0. Keluar
Pilih menu: 1
Nama Pembeli : y
No HP : 0826
Apakah antrian prioritas? (y/n): y
Antrian berhasil ditambahkan dengan nomor: 3

=====
SISTEM ANTRIAN ROYAL DELISH
=====
1. Tambah Antrian
2. Cetak Antrian
3. Hapus Antrian dan Pesan
4. Laporan Pesanan
0. Keluar
Pilih menu: 2

=====
Daftar Antrian Pembeli
=====
No Antrian Nama No HP
1 q 0821
2 w 0822
3 y 0826
4 e 0823
5 r 0824
6 t 0825
```